INCH-POUND

MIL-DTL-55302/69D
w/Amendment 1
6 March 2023
SUPERSEDING
MIL-DTL-55302/69D
24 March 2004

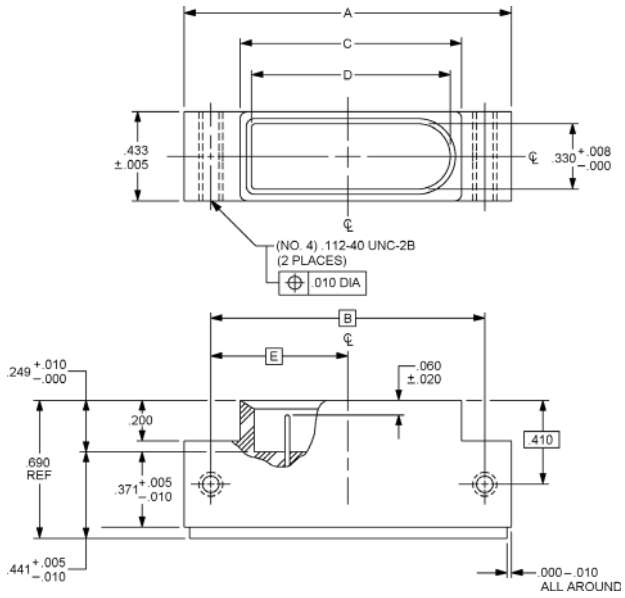
DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PIN PLUG, ELECTRICAL, POLARIZED (.090 SPACING) REMOVABLE CRIMP CONTACTS

MIL-DTL-55302/69D is inactive for new design after 1 October 1986.

This specification is approved for used by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.



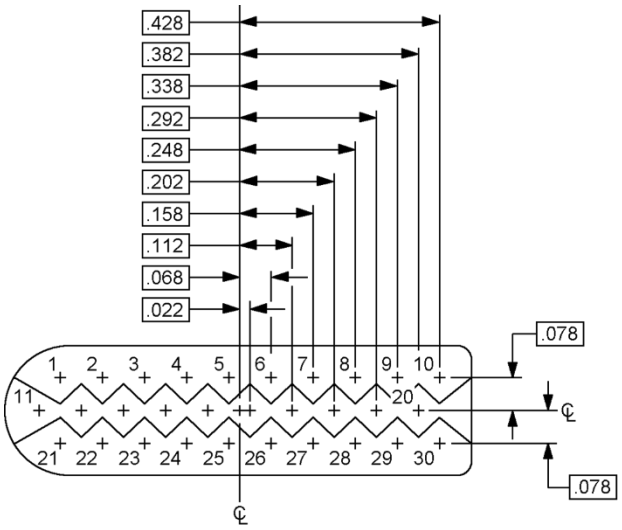
Inches	mm
.005	0.13
.008	0.20
.010	0.25
.020	0.51.
.060	1.52
.112	2.84
.200	5.08
.249	6.32
.330	8.38
.371	9.42
.410	10.41
.433	11.00
.441	11.20
.690	17.53

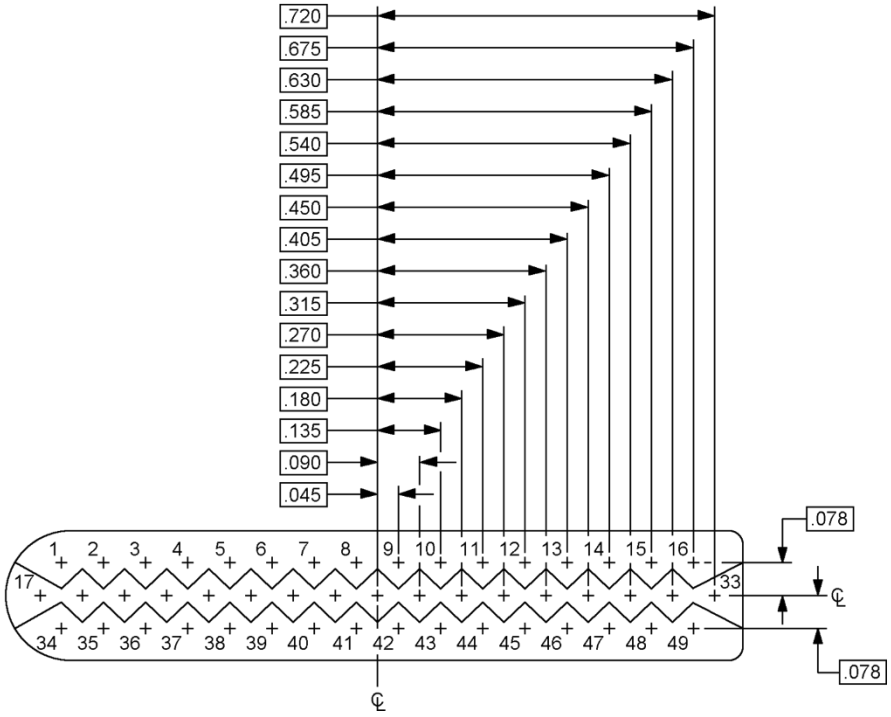
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Nominal spacing between any two adjacent pin contacts shall be .090 ±.003 (2.29 ±.08 mm).
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
5. For dimensions A through E see table I.
6. A quantity of M39029/58-354 socket crimp contacts consisting of the normal complement plus one spare contact for connector arrangements having 26 contacts or less and two spares for arrangements having over 26 contacts shall be supplied with each connector.

FIGURE 1. Connectors, plug, .090 (2.29 mm) contact spacing removable crimp pin.

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NOTES:
1. Contacts 1, 20 and 21 (top of page) are identified on the front face of the connector and contacts 1, 10, 11, 20, 21 and 30 are identified on the rear face per the insert arrangement.
2. Contacts 1, 16 and 49 (bottom of page) are identified on the front face of the connector and contacts 1, 16, 17, 33, 34 and 49 are identified on the rear face per the insert arrangement.

FIGURE 2. Insert arrangement for .090 (2.29 mm) contact spacing connector (male engaging face).

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Inches	mm	Inches	mm
.022	0.56	.292	7.42
.045	1.14	.315	8.00
.068	1.73	.338	8.59
.078	1.98	.360	9.14
.090	2.29	.382	9.70
.112	2.84	.405	10.29
.135	3.43	.428	10.87
.158	4.01	.450	11.43
.180	4.57	.495	12.57
.202	5.13	.540	13.72
.225	5.72	.585	14.86
.248	6.30	.630	16.00
.270	6.86	.675	17.15
		.720	18.29

Requirements:

Dimensions and configuration: See figure 1 and table I.

Material: In accordance with MIL-DTL-55302, except:

Body - reinforced epoxy resin or equivalent.

Bushing - monel, QQ-N-281, class B, cold drawn.

Plating: In accordance with SAE-AS39029/58.

Contact identification: Contact locations are identified numerically as shown on figure 2.

Mating and unmating: The maximum mating force, in pounds, shall be 0.5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of 0.08 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: With a test current of 3 A dc, the average resistance of all contact pairs measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:
Sea level: 1,000 volts rms, 60 Hz, ac.
High altitude: 275 volts rms, 60 Hz, ac.

Current rating: 3.0 amperes.

Insert arrangement: See figure 2.

Mating connectors: See MIL-DTL-55302/68.

Contact insertion and removal tool: See SAE-AS81969/8.

Quality conformance inspection:

Group A inspection: Delete mating and unmating.

Group B inspection: Add mating and unmating.

Part or identification number (PIN): M55302/69-(dash number form table I).

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TABLE I Dash number and dimensions.

Dash number	Dimensions 1/				
	+.010 A (.25) -.000	B	+.010 C (.25) -.000	+.008 D (.20) -.000	E
0 1	1.661 (42.19)	1.376 (34.95)	1.096 (27.84)	.996 (25.30)	.688 (17.48)
0 2	2.246 (5.7.05)	1.960 (49.78)	1.686 (42.82)	1.581 (40.16)	.980 (24.89)

1/ Metric equivalent are given for information only. Millimeters are in parentheses.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

In addition to MIL-DTL-55302, this specification sheet references the following documents.

SAE-AS39029/58
MIL-DTL-55302/68
SAE- AS81969/8
QQ-N-281

CONCLUDING MATERIAL

Custodians:
Air Force - 85
DLA - CC

Review activities:
Air Force - 19

Preparing activity:
DLA - CC

(Project 5935-2023-018)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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